

CULTIVARS

- Brandywine
- Cherokee Purple
- Green Zebra
- Roma
- San Marzano

INTENSIVE SPACING

- 1 row per bed, spaced every 24" (60 cm) on the row.

FERTILIZATION

- Field tomatoes are heavy feeders that take 100 days or more to ripen in the garden. We front-load the total amount of fertility requirement at the initial bed prep stage and do not top dress them afterwards.
- Apply lots of compost, +/- 1.5 inches (4 cm) all across the bed along with a dose of nitrogen-rich organic fertilizer such as pelleted chicken manure (5-3-2) or alfalfa meal (2-0-2). Also, add potassium sulphate (0-0-50) and organic Sul-Po-Mag (0-0-22)

IRRIGATION

Irrigation Type

- Drip tape (4 per bed) : under plastic mulch.
- No sprinkler to keep foliage dry and thus prevent diseases.

Irrigation Management

- Root depth : medium (24"-40" or 60-100 cm)
- General : constant and even moisture from flower through fruit. Field tomatoes like to dry down before being watered again.

STARTING SEEDLINGS

- Start transplants 7-8 weeks before the transplantation date. Sow seeds in 128 cell flats in seed-starting soil mix. 1 seed per cell.
- Place trays on heating mats and maintain floor temperature at 80 °F (27 °C) until emergence.
- After emergence, maintain the ambient temperature in the nursery around 75°F (24 °C) during the day and 65°F (18 °C) at night. The soil mix must be kept moist at all times.
- Pot-up in 4" (10 cm) pots 3 weeks after sowing. Use compost-rich repotting mix.
- Plants are ready to transplant when the seedlings are about 7-8 weeks old and 12-16" (30-40 cm) tall. Ideally, they should be transplanted before they start to flower.
- Harden off plants by exposing them to outside temperatures and reducing water for 4-5 days before transplanting.

IMPLANTATION

Field tomatoes are planted once a year. Like bell peppers and eggplants, they are heat-loving crops that like to be grown under cover (tunnels, caterpillars).

EQUIPEMENT

- Broad fork
- Rotary harrow
- 30" bed prep rake
- Landscape fabric, staples, drip tape and fittings.
- 5' (1.5 m) steel bars (cement rods); 1 stake every 4 plants.

TRANSPLANT ITINERARY

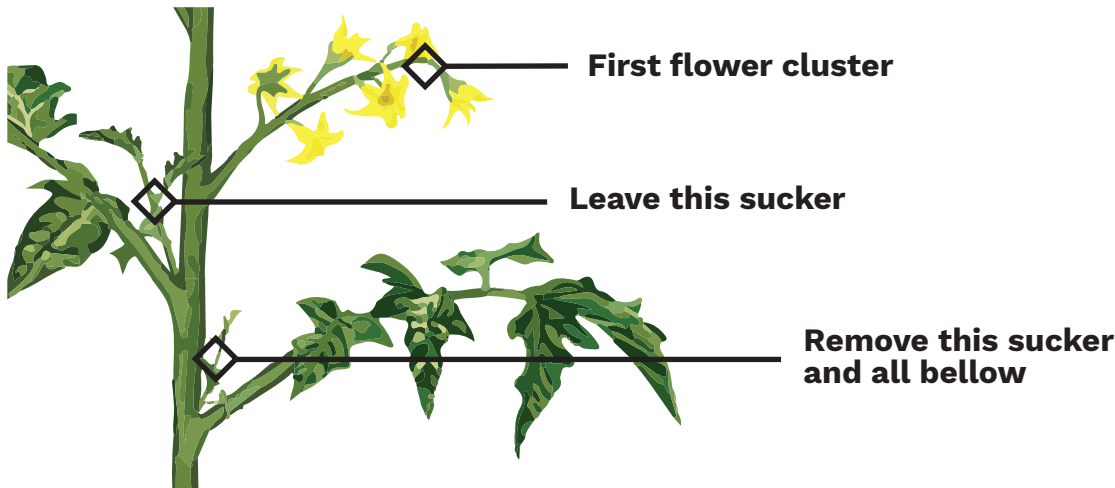
- Before transplanting, make sure soil temperature is above 60°F (16°C). Optimal soil temperature is 67°F (19°C) or higher. Covering the soil with clear plastic 2 weeks prior to the transplant date is a great way to increase soil temperatures.
- Prepare the soil with a pass of the broad fork.
- Remove large debris from the bed if needed (use flail mower, rake or manually remove the large chunks).

- Apply compost and amendments in the middle of the bed and do a pass with the rotary harrow at a depth of 2.5" (6 cm). After the bed is covered with landscape fabric, it does not need any raking.
- Install 4 lines of drip tape per bed and start the system. Fix any leaks at that point.
- Lay down landscape fabric and secure with staples each second step (every 5' or 1.5 m). It needs to be tight. The holes in the landscape fabric should be 5" (12 cm) in diameter to fit the 4" (10 cm) pots.
- Field tomatoes are not grafted and can, therefore, be planted deep in the ground. Aim to bury the plant up to the cotyledons or first branches. This will cause the plant to grow more roots, enhancing the overall fruit production and leading to early fruit maturity.
- To do this, use a trowel to dig a hole. Place the plant firmly in the bottom and fill all side cavities with the extra soil. Press firmly and use extra soil to bury the plant as much as possible.
- Make sure that both the plug and the soil are wet, as this greatly impacts rapid root uptake.
- Using a sledgehammer, drive the support stakes deep into the ground, 1 every 4 plants.
- Water the bed thoroughly the first time, keeping the soil moist for the first 2 weeks after the transplant.
- Once the plants have reached +/- 16" (40 cm) high, install the first line of bailing twine, crossing it from the inside out of each stake. Basket weave the plants 2 or 3 more times for every 10" (25 cm) they grow.

WEEDING

- Weed through the holes of the landscape fabric 2 or 3 times over the season.

MAINTENANCE



- We grow field tomatoes at 24" (60 cm) spacing, because we grow them on 2 stems. This optimizes both the cost of production and overall yields. The extra stem we keep is the one that develops from the sucker just below the first flower cluster. See diagram above.
- If possible, suckers should be removed at least once every other week.
- Prune leaves every couple of weeks, starting from the bottom up. This will improve air circulation and let some light inside the plant profile. Remove all damaged leaves. If the damaged is due to disease, remove leaves from the ground and throw them in the garbage.
- When pruning leaves, you should prune enough of the foliage to be able to see the colour of your colleague's shirt through the plants.
- 6 weeks prior to the last desired harvest, remove all growing tips on the tomato plants. This action will allow plants to finish ripening all the fruits that they have already formed. Make the cut above the last pollinated flower cluster.

PLANT PROTECTION

- Tomato plants are susceptible to many different diseases. *Cornell University's Vegetable MD Online has a Tomato Disease Identification Key* that can help you learn the symptoms of diseases, as well as how to distinguish between environmental or nutrient-related causes. Once you have identified a disease, use *Johnny's Selected Seeds Pest & Disease Chart* to find a recommended treatment.

HARVESTING

- Harvesting is conducted twice a week. Only the ripe fruits are harvested. All damaged or split fruit is set aside during harvest.
- The goal is to handle the tomatoes only once.

STORING

- Tomatoes are stored in a cold room kept at +/- 55F (10-12°C).